- The memory unit stores programs as well as input, output and intermediate data.
- The processor unit performs arithmetic and other data-processing tasks as specified by a program.
- The control unit supervises the flow of information between the various units.
- The control unit retrieves the instructions, one by one, from the program which is stored in memory.
- For each instruction, the control unit informs the processor to execute the operation specified by the instruction. Both program and data are stored in memory. The control unit supervises the program instructions, and the processor manipulates the data as specified by the program.

1.2 Binary Numbers

- A decimal number such as 7392 represents a quantity equal to 7 thousands, plus 3 hundreds, plus 9 tens, plus 2 units. The thousands, hundreds, etc. are powers of 10 implied by the position of the coefficients.
- To be more exact, 7392 should be written as: $7\times10^3+3\times10^2+9\times10^1+2\times10^0$
- However, the convention is to write only the coefficients and from their position deduce the necessary powers of 10. In general, a number with a decimal point is represented by a series of coefficients as follows:

$$\boxed{a_5 a_4 a_3 a_2 a_1 a_0 \, a_{-1} a_{-2} a_{-3}}$$

- The a_j coefficients are one of the ten digits $(0,1,2,\ldots,9)$, and the subscript value j gives the place value and, hence, the power of 10 by which the coefficient must be multiplied.

$$\boxed{10^5 a_5+10^4 a_4+10^3 a_3+10^2 a_2+10^1 a_1+10^0 a_0+10^{-1} a_{-1}+10^{-2} a_{-2}+10^{-3} a_{-3}}$$

- The decimal number system is said to be of **base**, or **radix**, 10 because it uses ten digits and the coefficients are multiplied by powers of 10.

Binary system

- The coefficients have two possible values: 0 and 1. Each coefficient a_j is multiplied by 2^j. For example, the decimal equivalent of the binary number 11010.11 is 26.75, as shown from the multiplication of the coefficients by powers of 2:

$$\boxed{1\times2^4+1\times2^3+0\times2^2+1\times2^1+0\times2^0+1\times2^{-1}+1\times2^{-2}=26.75}$$